

LATTICE-R: Trajectory-Aware, Layer-Routed Instruction-Tuning Data Selection from One Residualized Attribution, Certified by an Interventional Routing Test A Stage-1 Registered Report

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Abstract

Neuron-aware instruction-tuning data selection (NAIT) scores a candidate example by collapsing its decoder-layer activations to a per-layer principal direction, summing the layer projections into a single scalar, and selecting the top- k . That sum discards *which* layer carried the evidence and tunes every layer uniformly. We ask whether the per-layer axis NAIT already computes can be turned into a *decision variable* that drives both *which* examples to keep *and where* each example’s gradient should land—and whether doing so beats NAIT at a compute-matched budget. **LATTICE-R** derives, from one per-example endpoint+NAIT-residualized trajectory attribution $\psi(x) \in \mathbb{R}^{|\mathcal{A}|}$, two coupled readouts: the data-selection score is the *sum* $\sum_{\ell \in \mathcal{A}} \psi_{\ell}(x)$ and the LoRA layer-routing policy is the *support* $\text{supp}(\psi(x))$, tied by the exact identity $\sum_{\ell \in \mathcal{A}} \psi_{\ell}(x) = \hat{\beta}_T \cdot \tilde{\mathbf{T}}(x)$. We make the routing object a well-posed estimand—a *pool-conditional policy value* $V(\pi) = \mathbb{E}_{\text{seed}}[U_{\text{train}}(\pi) \mid \text{locked pools}]$ over whole training runs, so there is no per-example potential outcome and no cross-example SUTVA violation—and we pre-register an interventional routing test *that would certify* whether the coupling is load-bearing: six real masked-LoRA training arms, decided by an *intersection-union test* (each contrast at the marginal level, no multiplicity penalty). The Intersection-Union construction (Berger 1982) is exact *conditional on* valid level- α component tests; the implemented per-contrast inference is asymptotic (paired studentized bootstrap- t), so the finite-sample level is approximate. (These runs are not yet executed.) The headline claim is that LATTICE-R beats a *faithfully reproduced* NAIT comparator (its layer-sum scored over the full released decoder set) and every locked baseline on retention-adjusted target utility at a *measured* compute-matched budget. A ridge-FWL/double-ML mechanism certificate, computed out-of-sample through a frozen nuisance map, licenses the method non-circularly; a three-pool firewall, a measured compute ledger, and a closed-testing graph with an in-document FWER bound complete the plan. This is a *Stage-1 Registered Report*: method, theory, and analysis plan are locked *before* data collection. We report **no** experimental numbers; every results slot is an explicit placeholder. The v5 modules *are implemented* as additive do-not-run pure-Python with a green unit harness of **93 passing** formula-evaluation tests (`server.authorized=false`, no run executed); those are code-correctness checks, not evidence.

1 Introduction

The gap. Instruction tuning improves how language models follow instructions, but its returns depend heavily on *which* examples are used [45, 5]. The base method we build on, Neuron-Aware Data Selection (NAIT) [7], reads decoder-layer activations on in-domain data, reduces each layer

to a sign-aligned principal direction v_ℓ , scores a candidate by the *layer-sum projection* $s_{\text{NAIT}}(y) = \sum_{\ell=1}^L (A^{(\ell)}(y) \cdot v_\ell)$, and selects the top- k (base Algorithm 1, Eqs. 1–6). Two structural commitments are baked in. **(i) Layer collapse to a sum, then top- k :** the layer-sum destroys *which* layer carried the alignment evidence, and the method tunes *all* parameters uniformly. **(ii) Selection only:** NAIT answers “which examples,” never “given this example’s internal evidence, *where* should its gradient land.” The per-layer axis is computed and then discarded. The open question is whether keeping that axis as a decision variable—using one attribution to choose both the examples *and* the per-example tuning footprint—predicts downstream *utility* at a *compute-matched* budget, against a faithfully reproduced NAIT and a strong field of selectors that, recent work warns, often fail to beat random [9].

Why this is hard to do honestly. A naive “trajectory/routing features beat endpoints” comparison is easy to get wrong in four ways. **(a) Re-encoding:** a trajectory or routing summary can silently re-encode the endpoint signature, so a raw gain is an artifact of endpoint variance, not new information. We control the *full* endpoint geometry by ridge orthogonalization, not a low-rank shadow. **(b) An ill-posed routing estimand:** a per-example “turn layer ℓ off” contrast under fixed capacity is *not* a single potential outcome—freeing rank r_ℓ forces a redistribution with no canonical rule—and a shared LoRA run violates cross-example SUTVA. We therefore make the estimand a *policy value* over whole runs (the treated unit is the run, not the example). **(c) Invalid out-of-sample tests:** a nested-model F on out-of-sample residual sums of squares has no valid reference; “beats every control” tested with a shared min-over-controls (max- T) quantile is anti-conservative at the least-favorable configuration. We use a cross-fit conditional-null permutation for the mechanism certificate and an *intersection-union test* [1] for every “beats all” decision. **(d) Uncounted compute:** equal LoRA rank is not equal compute. We require a *measured* compute-matched ledger (params, optimizer slots, realized FLOPs, skip-flag) and forbid “savings” without a measured reduction.

Our approach (LATTICE-R). We fit one per-example endpoint+NAIT-residualized trajectory attribution $\psi(x) \in \mathbb{R}^{|\mathcal{A}|}$ over an anchor-layer set $\mathcal{A}$ ($|\mathcal{A}| = 8$) and read it *twice*: the data-selection score is the *sum* $\sum_{\ell \in \mathcal{A}} \psi_\ell(x)$ and the LoRA layer-routing *policy* π_ψ allocates a fixed rank budget R_{tot} over $\mathcal{A}$ to the *support* of $\psi(x)$, holding the non-anchor layers $\mathcal{L} \setminus \mathcal{A}$ at a fixed baseline rank r_0 shared by every arm. The coupling identity (Sec. 3)

$$\sum_{\ell \in \mathcal{A}} \psi_\ell(x) = \hat{\beta}_T \cdot \tilde{\mathbf{T}}(x)$$

ties the two readouts to *one* orthogonalized coefficient $\hat{\beta}_T$ on *one* out-of-sample residual $\tilde{\mathbf{T}}(x) = \mathbf{T}(x) - \mathbf{Z}(x)\mathbf{B}_\lambda$. We pre-register an *interventional routing test* (Gate R1) to certify the coupling once the authorized run executes: six real masked-LoRA arms over $\mathcal{A}$ (π_ψ , uniform, shuffled- ψ , random-mask, global-surgical, AdaLoRA), all compute-matched, decided by an intersection-union test of the pool-conditional policy-value gaps. The headline (Gate R2) is the pre-registered claim that LATTICE-R beats a faithful NAIT comparator scored over the full $\mathcal{L}$ and every locked baseline on retention-adjusted utility at a *measured* matched budget.

Position relative to NAIT, in one line. *NAIT sums the layer axis away (over all $\mathcal{L}$) and tunes uniformly; LATTICE-R keeps the layer axis as a per-example decision variable for selection and capacity allocation over $\mathcal{A}$, derives both from one endpoint+NAIT-residualized attribution, and is designed to test, under real matched-compute training runs that the authorized run will execute, whether the routing policy’s value $V(\pi_\psi)$ beats every prior-art arm (random, AdaLoRA, surgical,*

NeFT, MoE-routing) at a compute-matched budget—each by its own marginal- α margin bound. (Runs are not yet executed; this is the pre-registered test, not a demonstrated result.)

Falsifiable central claim.

One endpoint+NAIT-residualized trajectory attribution ψ drives both data selection (the sum) and LoRA layer routing (the support); at a measured compute-matched budget, the routing policy’s pool-conditional value $V(\pi_\psi)$ exceeds that of every capacity-matched control by margin δ_{R1} (each control beaten by its own marginal- α lower bound), and the resulting selection+routing method beats a faithfully reproduced NAIT (scored over the full $\mathcal{L}$) and every locked baseline on retention-adjusted target utility.

This is decisively falsifiable: if the residual attribution is null (Gate R0), or routing fails to beat every control (Gate R1), or the matched-budget win fails (Gate R2), the pre-registered fallback fires and the negative result is a clean, publishable falsification.

Contributions.

1. **A coupled selection-and-routing method from one attribution.** Selection = sum, routing = support of the *same* endpoint+NAIT-residualized attribution, tied by an exact identity (Sec. 3, Lemma 1). NAIT cannot express this: it discards the per-layer axis it computes.
2. **A well-posed, fully-specified routing estimand.** The routing object is a *pool-conditional policy value* $V(\pi)$ over whole runs (no per-example potential outcome, no SUTVA), with the policy domain pinned to $\mathcal{A}$, the mask→rank map a closed-form, deterministic, *score-free* largest-remainder apportionment, and every control a single deterministic map of x .
3. **An interventional routing certificate.** Six real masked-LoRA arms decided by an intersection-union test over a paired studentized bootstrap- t (no multiplicity penalty; valid at the least-favorable configuration). The Intersection-Union construction (Berger 1982) is exact *conditional on* valid level- α component tests; the implemented per-contrast inference is asymptotic (paired bootstrap- t), so the finite-sample level is approximate.
4. **A faithful NAIT comparator and an honest budget.** NAIT reproduced over the full released $\mathcal{L}$ (base Eq. 5), both base variants run, the stronger gating the headline; a measured compute ledger with an extraction-parity kill.
5. **Theory.** Strict propriety of the Brier factuality calibrator; dual-ridge feasibility/equivalence; residual orthogonality; the coupling identity; the intersection-union size bound; and the closed-testing FWER bound (Sec. 4).
6. **A locked, falsifiable protocol with a green implementation.** Datasets, models, baselines, multiplicity, margins, and kill-gates are pinned; the modules are implemented as do-not-run pure-Python with a green 93-test harness (`server.authorized=false`; Sec. 5).

Registered-Report status and implementation status. Under the Registered Report model [6], the unit of review is the question, method, and pre-registered analysis—not results. “No results yet” is the *expected* Stage-1 state. We present a complete method, theory, and locked plan; **empirical validation follows the authorized run**, and every results slot in Sec. 6 is a labeled placeholder. To make the plan executable and auditable now, the v5 modules *are already implemented* as additive, do-not-run, pure-Python (stdlib only; no model load, no server call, no training); they

are exercised by a green unit harness of **93 passing** formula-evaluation tests (Sec. 5, Table 1). `server.authorized` stays **false**; *no run is executed*, and these tests are code-correctness checks, not evidence.

2 Related Work

Position relative to the base paper (NAIT). NAIT [7] selects instruction examples by neuron-level activation alignment, collapsing each layer to a principal direction and *summing* the per-layer projections over the released decoder set. We do not claim neuron-aware selection is new: NAIT is both our must-beat comparator and our control. The endpoint signature ϕ_{end} and the faithful NAIT layer-sum s_{NAIT} (over $\mathcal{L}$) both enter our full control block, so the contribution lives entirely in the part of the attribution that survives partialling out NAIT’s own readout; by construction it cannot be an artifact of re-encoding the endpoint or NAIT score. Crucially, NAIT *scores* over the full $\mathcal{L}$ (base Eq. 5), so our faithful comparator does too; LATTICE-R *routes* only over the anchor set $\mathcal{A}$. These are different objects (comparator score vs. capacity mask) and there is no conflict.

Instruction-tuning data selection (and its threat model). Quality- and diversity-based selection [5, 45], importance resampling [40], and influence/gradient attribution [25, 33, 39] select examples without reading internal reasoning trajectories. LESS [39] and TracIn [33] attribute utility through gradients; we reuse an influence-on-validation attribution only to *define* our observational outcome Y_{obs} (so Y_{obs} is method-neutral), then ask whether a trajectory representation predicts that utility beyond endpoints, and whether *routing* on it helps. We adopt the threat model of [9]: selection strategies generalize poorly and often fail to beat random, with selection cost sometimes exceeding full fine-tuning. Random, full-data, zero-shot, and gradient-representation selectors are therefore all required baselines, and Gate R2 is gated on beating the *strongest* comparator, not NAIT alone. Coreset/diversity methods [35] are neighbours; we cite the classic core-set formulation rather than an unverified recent identifier.

Layer-/neuron-selective and rank-adaptive tuning (the routing prior art). AdaLoRA [44] reallocates LoRA rank *globally during training* by an SVD-importance score; Surgical Fine-Tuning [27] tunes one *fixed* layer block for a whole task; NeFT [41] tunes selected neurons, task-globally. LATTICE-R differs on two axes simultaneously: its allocation is *per example* (conditioned on that example’s internal evidence) and *a priori* (decided before training from the attribution), and it is *coupled* to the selection score. AdaLoRA, a global-surgical block, and (optionally) NeFT are required R1 routing arms, projected to the same integer capacity grid as every other arm, so that “routing on ψ beats global/adaptive rank allocation” is measured, not assumed.

Inference, distances, calibration, and submodular selection. We rely on partialling-out [13, 29], its modern double-ML form [8], and ridge regularization [21] for the residual estimand; permutation/bootstrap inference [16, 11, 19] and BCa intervals [10] for valid out-of-sample testing; the intersection-union test [1] for every “beats all” decision and a graphical sequentially-rejective scheme [3, 22] for cross-family gating; strictly proper scoring [4, 15] and calibration error [18] for the factuality gate; intraclass correlation [36] for outcome reliability; Fisher-based stability [24] for retention; the Nemhauser–Wolsey–Fisher $(1 - 1/e)$ guarantee [32] with lazy greedy [31, 38] for selection; sliced-Wasserstein distances [2, 26, 37] and kernel MMD [17] for the distributional term; and LoRA [23] for matched-budget tuning. Transformer-circuit analyses [12] motivate treating the

residual stream as a path. Our novelty is not any single tool but their composition into a coupled selection-and-routing method certified by a policy-value routing test.

3 Method

Dependency order: faithful NAIT (control + comparator) $\rightarrow$ trajectory operator $\mathbf{T}(x) \rightarrow$ endpoint+NAIT-residualized attribution ψ and the *coupling identity* $\rightarrow$ routing *policy* π_ψ over $\mathcal{A}$ with a fully-specified mask $\rightarrow$ rank map and its *pool-conditional* policy value $V(\pi) \rightarrow$ compute-matched method comparison. Two outcome symbols are kept distinct and *never* conflated: $Y_{\text{obs}}(x)$ is the observational per-example utility (the R0 regression target that ψ is fit to predict), and $U_{\text{train}}(\pi)$ is the held-out *training-outcome* functional (what R1/R2 score). The implemented modules realizing each block are named inline and listed in Table 1.

3.1 Notation and the faithful NAIT baseline

For a transformer with released decoder set $\mathcal{L}$ (the NAIT base model is LLaMA-2-7b, so $|\mathcal{L}| = 32$) and hidden width d , write the residual-stream activation at token position p , layer ℓ as $h_\ell(p) \in \mathbb{R}^d$. Over the anchor set $\mathcal{A} \subseteq \mathcal{L}$ ($|\mathcal{A}| = 8$, fixed on validation), the endpoint signature is

$$\phi_{\text{end}}(x) = \text{concat}_{\ell \in \mathcal{A}} [h_\ell(p_{\text{start}}), h_\ell(p_{\text{end}})] \in \mathbb{R}^{2d|\mathcal{A}|}. \quad (1)$$

Faithful NAIT over the full $\mathcal{L}$ (`baselines/nait_layerwise.py`; base Eqs. 2–6). For every $\ell \in \mathcal{L}$ form $\Delta A^{(\ell)}(P_i) = A^{(\ell)}(t_K) - A^{(\ell)}(t_1)$ (Eq. 2), $v_\ell = \text{PCA}_1(\{\Delta A^{(\ell)}(P_i)\}_i)$ (Eq. 3), sign-aligned by $v_\ell \leftarrow -v_\ell$ if $\mu_{\text{diff}} \cdot v_\ell < 0$ (Eq. 4), and score

$$s_{\text{NAIT}}(y) = \sum_{\ell=1}^L (A^{(\ell)}(y) \cdot v_\ell), \quad \text{select} = \text{top-}k(s_{\text{NAIT}}). \quad (2)$$

Both base variants (first/last difference, Eq. 2; and the prose token-mean summary) are reproduced because the base paper is internally inconsistent; the *stronger* is the comparator. A restricted 8-anchor score $s_{\text{NAIT}}^A = \sum_{\ell \in \mathcal{A}} (A^{(\ell)} \cdot v_\ell)$ is a *secondary, gated* diagnostic only, reported after the full- $\mathcal{L}$ reproduction passes its unit check; it is *never* the comparator the headline uses. The endpoint reconstruction (`baselines/nait.py`, ϕ_{end}) is retained as a *distinct* full endpoint control that must not coincide with the layerwise sum.

3.2 Trajectory operator

Let $H(x) = \{h_{\ell,s,t}\}$ index layers ℓ , decoding steps s , tokens t .

Distributional term D_ℓ (higher moments). Collect the per-layer cloud $P_\ell(x) = \{h_{\ell,s,t}\}$ (subsampled to a persisted cap of 512 vectors) and compare it to a held-out target cloud Q_ℓ^T by the sliced-Wasserstein-2 distance, estimated with $K=64$ persisted random projections:

$$D_\ell(x) = \text{SW}_2^2(P_\ell(x), Q_\ell^T), \quad \text{SW}_2^2(P, Q) = \mathbb{E}_{u \sim \text{Unif}(S^{d-1})} [W_2^2(u_\# P, u_\# Q)]. \quad (3)$$

Temporal curvature κ_ℓ (uses ordering). With per-step mean path $g_{\ell,s} = \text{mean}_t h_{\ell,s,t}$, velocity $v_{\ell,s} = g_{\ell,s+1} - g_{\ell,s}$, acceleration $a_{\ell,s} = g_{\ell,s+1} - 2g_{\ell,s} + g_{\ell,s-1}$, the normalized bending energy is

$$\kappa_\ell(x) = \frac{1}{S_x - 2} \sum_{s=2}^{S_x-1} \frac{\|a_{\ell,s}\|}{\|v_{\ell,s}\| \|v_{\ell,s-1}\| + \varepsilon}. \quad (4)$$

The cross-layer signature stacks the per-layer coordinates over $\mathcal{A}$:

$$\mathbf{T}(x) = (\{D_\ell(x)\}_{\ell \in \mathcal{A}}, \{\kappa_\ell(x)\}_{\ell \in \mathcal{A}}) \in \mathbb{R}^{2|\mathcal{A}|}. \quad (5)$$

3.3 Endpoint+NAIT-residualized attribution and the frozen out-of-sample map

Define the per-example *control feature row* for *any* x (in or out of sample),

$$\mathbf{Z}(x) = [\phi_{\text{end}}(x), s_{\text{NAIT}}(x), V_{\text{proj}}(x), \mathbf{C}(x), 1] \in \mathbb{R}^p, \quad (6)$$

where $V_{\text{proj}}(x) \in \mathbb{R}^{|\mathcal{L}|}$ collects the per-layer NAIT projections $A^{(\ell)}(x) \cdot v_\ell$ (summing to s_{NAIT}) and $\mathbf{C}$ are covariates (length, difficulty, family one-hots). Penalizing only the wide $[\phi_{\text{end}}, V_{\text{proj}}]$ blocks via $\Omega = \text{blockdiag}(\mathbf{I} \text{ on } [\phi_{\text{end}}, V_{\text{proj}}], \mathbf{0} \text{ on } [s_{\text{NAIT}}, \mathbf{C}, 1])$, we fit *once* on the training fold and *freeze* the ridge nuisance coefficient maps (`analysis/layer_attribution.py::frozen_nuisance_map`),

$$\mathbf{B}_\lambda = (\mathbf{Z}^\top \mathbf{Z} + \lambda \Omega)^+ \mathbf{Z}^\top \mathbf{T} \in \mathbb{R}^{p \times 2|\mathcal{A}|}, \quad \mathbf{b}_\lambda^Y = (\mathbf{Z}^\top \mathbf{Z} + \lambda \Omega)^+ \mathbf{Z}^\top Y_{\text{obs}}. \quad (7)$$

The deployed, *out-of-sample* residuals for any x are closed-form functions of x through its own row $\mathbf{Z}(x)$ (`residualize_out_of_sample`):

$$\tilde{\mathbf{T}}(x) = \mathbf{T}(x) - \mathbf{Z}(x) \mathbf{B}_\lambda, \quad \tilde{Y}(x) = Y_{\text{obs}}(x) - \mathbf{Z}(x) \mathbf{b}_\lambda^Y. \quad (8)$$

On the stacked training rows this reproduces the in-sample residual maker $\tilde{\mathbf{T}} = (\mathbf{I} - \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z} + \lambda \Omega)^+ \mathbf{Z}^\top) \mathbf{T}$ exactly, so no in-sample math changes; the only addition is that $\tilde{\mathbf{T}}(x)$ is now defined for rows $\mathbf{Z}$ never saw (Theorem 2). The orthogonalized trajectory coefficient (dual / n -space form, feasible at 7B width) on the observational target is

$$\hat{\beta}_T = (\tilde{\mathbf{T}}^\top \tilde{\mathbf{T}})^+ \tilde{\mathbf{T}}^\top \tilde{Y}, \quad (9)$$

and the registered residual quantity is the partial- R^2 , partial- $R_T^2 = (\text{RSS}_{\text{red}} - \text{RSS}_{\text{full}}) / \text{RSS}_{\text{red}}$, with the overall increment $\Delta R_{\text{overall}}^2$ on the TSS denominator mapped to the deployment margin.

Per-layer policy score and the coupling identity. The per-layer *routing policy score* is the per-layer contribution of $\hat{\beta}_T$ to the out-of-sample residual (`per_layer_policy_score`):

$$\psi_\ell(x) = \hat{\beta}_{D,\ell} \tilde{D}_\ell(x) + \hat{\beta}_{\kappa,\ell} \tilde{\kappa}_\ell(x), \quad \ell \in \mathcal{A}, \quad (10)$$

where $\tilde{D}_\ell, \tilde{\kappa}_\ell$ are the per-layer coordinates of $\tilde{\mathbf{T}}(x)$.

Lemma 1 (Coupling identity). *Because $\mathbf{T}(x)$ stacks the per-layer coordinates and $\hat{\beta}_T$ is the single orthogonalized coefficient on $\tilde{\mathbf{T}}$,*

$$\sum_{\ell \in \mathcal{A}} \psi_\ell(x) = \sum_{\ell \in \mathcal{A}} (\hat{\beta}_{D,\ell} \tilde{D}_\ell(x) + \hat{\beta}_{\kappa,\ell} \tilde{\kappa}_\ell(x)) = \hat{\beta}_T \cdot \tilde{\mathbf{T}}(x),$$

for out-of-sample x , since both sides use the same frozen $\mathbf{B}_\lambda, \hat{\beta}_T$.

Hence the **selection score** $\sum_{\ell \in \mathcal{A}} \psi_\ell(x)$ is exactly the endpoint+NAIT-residualized trajectory prediction $\hat{\beta}_T \cdot \tilde{\mathbf{T}}(x)$, and the **routing support** is $\text{supp}(\psi(x)) \subseteq \mathcal{A}$: selection is the *sum*, routing is the *support*, of *one* attribution. The identity is unit-tested in- and out-of-sample (`test_layer_routing.py`, item (a)). We emphasize ψ is a *predictive* policy score, not a causal layer-write effect; the certification that writing to $\text{supp}(\psi)$ is the right place is the policy-value Gate R1, not this decomposition.

3.4 The routing policy and the fully-specified mask→rank map

A routing *policy* maps each selected x to an anchor mask and integer ranks (`layer_function.py`):

$$\pi : x \mapsto \left(m_{\mathcal{A}}(x) \in \{0, 1\}^{\mathcal{A}}, \{r_{\ell}(x)\}_{\ell \in \mathcal{A}} \right), \quad \sum_{\ell \in \mathcal{A}} r_{\ell}(x) = R_{\text{tot}}, \quad r_{\ell}(x) = r_0 \quad \forall \ell \in \mathcal{L} \setminus \mathcal{A}, \quad (11)$$

so the $|\mathcal{L} \setminus \mathcal{A}|$ non-anchor layers form a fixed common substrate at rank r_0 that *cancels in every contrast*. The mask→rank map `capacity_match` is a *pure function of* $(m_{\mathcal{A}}, R_{\text{tot}}, r_{\text{max}})$ and *nothing else*—it reads no ψ —defined on the feasible domain $\text{DOM} = \{m_{\mathcal{A}} : |\text{supp}(m_{\mathcal{A}})| \cdot r_{\text{max}} \geq R_{\text{tot}}\}$. Given a feasible mask with support S of size k , it is the largest-remainder (Hamilton) apportionment: even base $q = \lfloor R_{\text{tot}}/k \rfloor$ on every ON layer, a +1 top-up of the $\text{rem} = R_{\text{tot}} - kq$ leftover units to the lowest-index ON layers (deterministic ascending- ℓ tie-break), and a per-layer cap r_{max} enforced among the same ON layers (no support growth). Feasibility is guaranteed *upstream* by the single constructor `make_feasible_mask(raw, score, Rtot, rmax)`, the *only* place an arm-specific score is read: it lifts the empty mask to uniform-over- $\mathcal{A}$ and completes any under- k_{min} mask ($k_{\text{min}} = \lceil R_{\text{tot}}/r_{\text{max}} \rceil$) with the highest-score OFF anchors. Thus `capacity_match` has *no infeasible branch* and is total on DOM; with r_{max} pinned so $|\mathcal{A}| \cdot r_{\text{max}} \geq R_{\text{tot}}$ (Sec. 5), $U_{\text{train}}(\pi)$ is one implementation-independent functional. π_{ψ} instantiates the raw mask $1[\psi_{\ell}(x) \geq \tau_{\text{sel}}]$, lifts it via `make_feasible_mask` (score = ψ), then applies `capacity_match`.

Lemma 2 (Feasibility / no-overflow; `capacity_match` is total and conserving on DOM). *For every $m_{\mathcal{A}} \in \text{DOM}$, $q = \lfloor R_{\text{tot}}/k \rfloor \leq R_{\text{tot}}/k \leq r_{\text{max}}$; after the even-base and largest-remainder steps every ON layer carries q or $q+1$. If $q < r_{\text{max}}$ then $q+1 \leq r_{\text{max}}$ and the cap step is a no-op; if $q = r_{\text{max}}$ then $R_{\text{tot}} = kr_{\text{max}}$ and $\text{rem} = 0$, so again no layer exceeds the cap. In both cases $\sum_{\ell} r_{\ell} = R_{\text{tot}}$, no support is grown, and the output is a deterministic function of $(m_{\mathcal{A}}, R_{\text{tot}}, r_{\text{max}})$.*

3.5 The pool-conditional policy value and the kill-gates

The held-out training-outcome functional is the retention-adjusted target utility of the model obtained by training the selected set under per-example masks+ranks $\{\pi(x)\}$ (with the fixed r_0 substrate on $\mathcal{L} \setminus \mathcal{A}$), evaluated on the frozen held-out split. The *policy value* is the *pool-conditional* seed-expectation, and the confirmatory estimand is the per-control gap:

$$V(\pi) = \mathbb{E}_{\text{seed}}[U_{\text{train}}(\pi) \mid P_{\text{train}}, P_{\text{val}}, P_{\text{dep}}], \quad g_c = V(\pi_{\psi}) - V(\pi_c). \quad (12)$$

The expectation is over training seeds only; the three pools are held at their single locked partition (Sec. 5). The treated unit is the whole run, so there is **no per-example potential outcome and no cross-example SUTVA**. The seed-mean $\hat{V}(\pi) = S^{-1} \sum_s U_{\text{train},s}(\pi)$ is unbiased for $V(\pi)$, and the bootstrap- t below resamples exactly the seed randomness it averages, so estimand and estimator are aligned and the claim is scoped to the locked pools.

Gate R0 (mechanism certificate; analysis/residual_test.py). Reject “**T** adds nothing given $[\phi_{\text{end}}, \text{NAIT}(\mathcal{L})]$ ” iff the conditional-null block permutation on the orthogonalized residuals $\tilde{\mathbf{T}}$ (within family×fold strata; $P=5000$; frozen cross-fit nuisance, $K=10$; no ridge refit inside a permutation) gives $p < \alpha_{R0}$ and the cluster BCa-95% CI of partial- R_T^2 excludes the floor, stable across $r \in \{8, 16, 32\}$ PCA poles. *Fail $\Rightarrow$ the method reduces to NAIT; report a clean null (stop_main_novelty_claim).*

Gate R1 (interventional routing certificate; analysis/routing_intervention.py). On P_{val} , six real masked-LoRA arms over $\mathcal{A}$ (shared r_0 , shared `capacity_match`, shared seeds, compute-matched): π_ψ ; π_{unif} (all anchors ON); π_{shuf} (a fixed within-stratum permutation of the anchor labels applied to ψ); π_{rand} (a deterministic pseudo-random subset of $\mathcal{A}$ matched in cardinality to π_ψ); π_{global} (a fixed global block A_{glob} , surgical-style); π_{ada} (AdaLoRA importance from a fixed warm-up). **Every control is a single deterministic map of x :** all randomness is driven by fixed, pre-registered seeds combined with the example’s stable content hash $H(x)$, so the randomization is *part of the policy definition, not part of the $V(\pi)$ training-seed expectation*; π_{ada} ’s importance is frozen by a *separate* seed seed_{ada} run once *outside* that expectation. The composite “ π_ψ beats every control by δ_{R1} ” is an intersection-union alternative,

$$H_0^{R1} = \bigcup_c \{g_c \leq \delta_{R1}\}, \quad H_1^{R1} = \bigcap_c \{g_c > \delta_{R1}\}, \quad (13)$$

decided by the intersection-union test [1]: reject iff *each* per-control one-sided lower bound $L_c(\alpha)$ exceeds δ_{R1} , every L_c from a paired studentized bootstrap- t (Sec. 3.6), at the *marginal* level α —*no* penalty across the five controls, *no* shared min-over-controls quantile. The reported simultaneous lower bound is $L_{R1} = \min_c L_c(\alpha)$. *Fail* $\Rightarrow$ *drop routing, keep selection-only* (`drop_routing_keep_selection`).

Gate R2 (compute-matched method win; analysis/matched_budget.py). At a *measured* compute-matched budget (`cost_model.py` ledger), LATTICE-R’s U_{train} must beat the stronger full- $\mathcal{L}$ NAIT variant *and* every baseline by the locked margins. The R2-target “beats every comparator” claim is, like R1, an intersection-union test over the baseline set (δ_{target}). The remaining sub-claims are single margin tests: a *lower* bound above δ_{rel} for the relative-win margin, and an *upper* bound below the ceiling for the non-inferiority drift/cost margins $\delta_{\text{ret}}, \delta_{\text{hall}}, \delta_{\text{cost}}$; *every* sub-claim must pass. *Fail* $\Rightarrow$ *no method-win claim* (`no_method_win_claim`).

Gate R3 (honest two-sided cost). `cost_model.py` reports the extraction-parity multiplier ($> 2.0 \times \Rightarrow$ high-cost-analysis) and the training-side ledger; a “training savings” claim is licensed *only* if a measured FLOP/wall-clock reduction with masked adapters actually skipped (skip-flag true) is recorded.

Descriptive per-layer notion (not causal). If a per-layer summary is desired, it is a run-level leave-one-anchor-layer policy-value difference $\Delta_\ell^{\text{LOL}} = V(\pi_{\text{full}}) - V(\pi_{-\ell})$ with the explicit `capacity_match` redistribution, labeled *descriptive, not causal evidence*, and *outside* the confirmatory α . The off-manifold layer-ablation profile $J_{c,\ell}$ may enter the router prior only if it passes a validation gate against Δ_ℓ^{LOL} ; else it is dropped and the router uses ψ alone.

3.6 Valid inference for the mechanism certificate and the “beats all” maxima

R0 (cross-fit conditional-null permutation). For each of $K=10$ family-stratified folds, fit $\mathbf{B}_\lambda, \mathbf{b}_\lambda^Y, \lambda$ on the other nine and form the out-of-fold residuals; permute the rows of $\mathbf{T}$ relative to $\tilde{Y}$ within family $\times$ fold strata, recompute the held-out partial- R_T^2 , and report the add-one p -value; bracket by cluster BCa ($B=2000$). No F -distribution p -value is ever used inferentially.

Per-contrast inference (R1/R2): paired studentized bootstrap- t . For control/comparator c , with per-seed paired differences $d_{c,s} = U_{\text{train},s}(\pi_\psi) - U_{\text{train},s}(\pi_c)$, gap $\hat{g}_c = \text{mean}_s d_{c,s}$, seed standard error se , resample seeds with replacement, recompute $t_c^{*(b)} = (\hat{g}_c^{*(b)} - \hat{g}_c)/\text{se}^{*(b)}$, and form the one-sided lower bound

$$L_c(\alpha) = \hat{g}_c - t_{c,1-\alpha}^* \text{se}. \quad (14)$$

This is *asymptotically valid* [19]; we claim *only* asymptotic validity (an earlier exact-finite- S sign-flip claim is withdrawn—i.i.d. seeds make the $d_{c,s}$ i.i.d. but not symmetric about their mean). The “beats all” decision is the IUT of Eq. (13): reject iff $L_c(\alpha) > \delta$ for every c , with $L_{R1} = \min_c L_c(\alpha)$ (and identically for R2-target).

4 Theory

These results are discharged at the design stage (they are mathematical properties of the estimators, independent of data); the corresponding numeric sanity checks (the Brier identity, the coupling identity, capacity conservation, the IUT size at the least-favorable configuration, submodularity) are *formula evaluations in the green unit harness, not evidence*.

4.1 Strict propriety of the Brier factuality calibrator

Theorem 1 (Strict propriety). *Score the calibrator output $q \in [0, 1]$ against a label $y \sim \text{Bernoulli}(p)$ by the Brier rule $\ell_{\text{Brier}}(q, y) = (q - y)^2$. The population risk $\mathbb{E}[(q - y)^2] = (q - p)^2 + p(1 - p)$ is uniquely minimized at $q = p$.*

Proof. $p(q - 1)^2 + (1 - p)q^2 = q^2 - 2pq + p = (q - p)^2 + p(1 - p)$; the term $p(1 - p)$ is independent of q and $(q - p)^2 \geq 0$ with equality iff $q = p$; taking expectation over contexts preserves strictness. $\square$

Propriety proves *calibration*, not eval-drift prediction; the factuality gate G6 therefore separately requires rank/linear correlation with held-out drift and an ECE ceiling before λ_f may be non-zero (else $\lambda_f := 0$). None of this enters the confirmatory routing α .

4.2 Dual feasibility of the frozen residual maker

Theorem 2 (Dual equivalence and out-of-sample feasibility). *Fix a train fold with penalized endpoint/NAIT design Φ , unpenalized design $\mathbf{U} = [s_{\text{NAIT}}, \mathbf{C}, 1]$, target y , penalty $\lambda > 0$. The ridge fit $\min_{a,b} \|y - \Phi a - \mathbf{U}b\|^2 + \lambda \|a\|^2$ has the same fitted values—hence the same coefficient map $\mathbf{B}_\lambda$ of Eq. (7)—as the dual system with $a = \Phi^\top \alpha$, $\mathbf{K} = \Phi\Phi^\top + \lambda\mathbf{I}$: $(\mathbf{U}^\top \mathbf{K}^{-1} \mathbf{U})b = \mathbf{U}^\top \mathbf{K}^{-1} y$, $\alpha = \mathbf{K}^{-1}(y - \mathbf{U}b)$. The dual touches the endpoint/NAIT width only through the $N \times N$ Gram; its cost is $O(N^2 p + N^3)$, independent of $2d|\mathcal{A}| + |\mathcal{L}|$, so the full control is feasible at $7B$ width and $\mathbf{B}_\lambda$ applies to any out-of-sample row.*

Proof. Stationarity in a gives $a \in \text{row}(\Phi)$, $a = \Phi^\top \alpha$ and $(\Phi\Phi^\top + \lambda\mathbf{I})\alpha = y - \mathbf{U}b$; stationarity in b is the unpenalized normal equation; eliminating α yields the Schur-reduced b -equation. Both solve the same strictly convex program ($\lambda > 0$), so the fitted values coincide; the Gram is the only object scaling with the control width, and $\mathbf{B}_\lambda$ is a fixed $p \times 2|\mathcal{A}|$ matrix evaluable on any $\mathbf{Z}(x)$. $\square$

4.3 Residual orthogonality

Lemma 3 (The residual estimand is orthogonal to the endpoint/NAIT geometry). *With $\tilde{\mathbf{T}} = \mathbf{T} - \mathbf{Z}\mathbf{B}_\lambda$ and $\tilde{Y} = Y_{\text{obs}} - \mathbf{Z}\mathbf{b}_\lambda^Y$, as $\lambda \rightarrow \infty$ on the penalized block $\hat{\beta}_T$ measures only the association between Y_{obs} and $\mathbf{T}$ orthogonal to $[\phi_{\text{end}}, s_{\text{NAIT}}, V_{\text{proj}}, \mathbf{C}, 1]$, and partial- $R_T^2 = 0$ whenever $\mathbf{T}$ is a linear function of that control block.*

Proof sketch. By Frisch–Waugh–Lovell [13, 29], the joint coefficient of $\mathbf{T}$ equals that from regressing $\tilde{Y}$ on $\tilde{\mathbf{T}}$; ridge on the penalized block is FWL with a shrinkage operator [21], and the unpenalized block is projected out exactly. If $\mathbf{T} = [\cdot]M$, then $\tilde{\mathbf{T}} \rightarrow \mathbf{0}$ on the spanned directions and the partial- R^2 vanishes. $\square$

This is the formal content of “beyond NAIT”: because s_{NAIT} and the per-layer V_{proj} are *in* the control, a residual NAIT’s own layer-sum explains cannot pass R0.

4.4 First-moment pooling is provably insufficient

Proposition 1 (Curvature is not a function of the first-moment pool). *There exist step-sequences with identical per-layer first-moment summaries but different κ_ℓ .*

Proof. A reordering of the steps leaves the activation-multiset mean invariant but changes the ordered second differences $a_{\ell,s}$ of Eq. (4) whenever the path is non-collinear, so κ_ℓ is not a function of the first-moment pool. NAIT’s first/last difference and token-mean summaries are exactly such first-moment summaries. $\square$

4.5 The intersection-union test is exact conditional on valid component tests

Proposition 2 (IUT size; Berger 1982). *Let $H_0 = \bigcup_{c=1}^J \{g_c \leq \delta\}$ and let each component reject iff $L_c(\alpha) > \delta$ with $L_c(\alpha)$ a level- α lower confidence bound for g_c . The IUT rejects H_0 iff all J components reject. Then for any $\theta \in H_0$ there is c^* with $g_{c^*} \leq \delta$, and $P_\theta(\text{reject}) \leq P_\theta(L_{c^*}(\alpha) > \delta) \leq P_\theta(L_{c^*}(\alpha) > g_{c^*}) \leq \alpha$. Hence the IUT has size $\leq \alpha$ with no multiplicity correction and for every J , regardless of dependence among contrasts. The supremum is at the least-favorable configuration ($g_{c^*} = \delta$, others $\rightarrow \infty$)—exactly where a min-over-controls shared quantile is anti-conservative.*

The Intersection-Union construction is therefore *exact conditional* on each $L_c(\alpha)$ being a valid level- α component bound. Our implemented $L_c(\alpha)$ is the per-contrast bootstrap- t bound [19], which is only *asymptotically* valid (i.i.d. seeds make the $d_{c,s}$ i.i.d. but not symmetric, so no finite- S exactness is available); the IUT thus inherits asymptotic, not finite-sample, validity, and the finite-sample level is approximate. What the union bound *does* guarantee unconditionally is that no multiplicity penalty is incurred for any J under arbitrary dependence among contrasts. This is unit-checked: the per-contrast test attains nominal level on a null DGP, and the IUT holds size $\leq \alpha$ at the single-binding-control configuration when fed valid component bounds (`test_layer_routing.py`, item (c)).

4.6 Closed-testing FWER on the corrected leaves

Theorem 3 (FWER bound). *On the confirmatory family $\{H_0^{R0}, H_0^{R1}, H_0^{G2t}, H_0^{G2r}, H_0^{G2h}, H_0^{G2c}\}$, with the pinned graphical scheme [3] $R0 \rightarrow R1 \rightarrow G2t \rightarrow \{G2r, G2h, G2c\}$ (weights $w_0=1$, edges 1.0, split (0.34, 0.33, 0.33), recycle 1.0) and the union-null leaves $R0, R1, G2t$ each tested in every intersection that contains them by the full-union IUT (Prop. 2) and each drift/cost singleton by the asymptotic bootstrap- t at its weighted-Holm α , the closed test controls the family-wise error rate at $\alpha = 0.05$ (asymptotically).*

Proof sketch. Each leaf local test holds its level: R0 by the frozen-nuisance within-stratum exchangeability of the permutation; every union-null leaf by the full-union IUT, whose size is $\leq$ its node α for the full union null regardless of the other members of the intersection (Prop. 2); each single margin leaf by the bootstrap- t . A *sub-scoped* IUT (a proper subset of a union’s components) is forbidden because it tests a strictly smaller hypothesis and is anti-conservative when the binding component is omitted. By the closed-testing principle, an elementary H_i is rejected only if all intersections containing i reject at their level; requiring the full union inside every such intersection only shrinks the rejection region, so $P(\text{reject any true } H_0) \leq \alpha$. The data-dependent “max over arms” cannot leak type-I error because the IUT requires *every* arm to clear the margin. $\square$

The shortcut/brute-force equivalence over all $2^6 - 1 = 63$ intersections, including the IUT leaves and the least-favorable single-binding-control configuration, is asserted in code (`analysis/closed_testing.py`; `test_layer_routing.py`, item (f)).

4.7 Selection objective: monotone-submodular $(1 - 1/e)$ guarantee

Theorem 4 (Greedy approximation). *With per-example utility $u(x) = \max(0, \sum_{\ell \in \mathcal{A}} \psi_\ell(x) - \lambda_r \hat{r}(x) - \lambda_f \hat{f}(x))$ and the facility-location coverage $C(S) = \sum_e w_e \max_{x \in S} \sigma(x, e)$, the objective $F(S) = \sum_{x \in S} u(x) + \mu C(S)$ ($u \geq 0, w_e \geq 0, \sigma \in [0, 1], \mu \geq 0$) is monotone submodular; greedy returns $F(S_{\text{greedy}}) \geq (1 - 1/e)F(S^*)$, and lazy greedy returns the identical set.*

Proof. $\sum_{x \in S} u(x)$ is modular and monotone; each $\max_{x \in S} \sigma(x, e)$ is monotone with non-increasing marginals (submodular); a non-negative weighted sum preserves monotone-submodularity, so the Nemhauser–Wolsey–Fisher bound [32] applies and lazy greedy [31] gives the same set. Routing is applied *after* selection, so it never enters F and the guarantee is untouched. $\square$

5 Experimental Setup (Pre-Registered Analysis Plan)

`server.authorized` stays `false`. Everything below is locked *before* data collection; *no command is executed here*. **All pre-registered decision thresholds—margins** ($\delta_{R1}, \delta_{\text{target}}, \dots$), **capacity** ($R_{\text{tot}}, r_{\text{max}}, r_0$), **floors** (`floor_partial`, ρ_J), **control seeds**, and **the closed-test α -allocation—are pinned NOW as design choices** (concrete values in the “Locked symbols, thresholds, and seeds” paragraph below and in `configs/experiments/lattice_v5.yaml1`); the multiplicity-structure constants (graph weights, edges, split, IUT family) are likewise pinned. Only genuinely *empirical* quantities (resolved commit/tokenizer hashes, measured FLOPs, and every observed result cell) remain `DATA_NEEDED`, resolved at run authorization.

Backbones (pinned, released checkpoints). Primary Qwen2.5-7B-Instruct; secondary Qwen2.5-1.5B-Instruct (two-size, Apache-2.0) [34]; alternates Llama-3.1-8B-Instruct / Mistral-7B-Instruct-v0.3. LoRA only [23]; matched rank/steps/tokens across all methods. The faithful NAIT comparator is reproduced over the released decoder set $\mathcal{L}$ of each backbone (the NAIT base model is LLaMA-2-7b, $|\mathcal{L}| = 32$). Exact commit revisions and tokenizer hashes are resolved (not invented) at run authorization.

Datasets (pinned identities; content hashes resolved at authorization). Candidate pool by family: MetaMathQA (math) [43], the permissive `glaiveai/glaive-code-assistant` code-instruct set [14], and HotpotQA-distractor (multi-hop) [42]; held-out retention eval MMLU [20]; hallucination eval TruthfulQA [28] + FActScore-style atomic-claim factuality [30] (also the factuality-gate ground truth).

Three-pool leakage firewall (analysis/pool_firewall.py). A persisted hash partitions the corpus into example-disjoint, family-stratified P_{train} (learn ψ , $\mathbf{B}_\lambda$, $\mathbf{b}_\lambda^Y$, NAIT directions, LOCI centroids; Y_{obs} here), P_{val} (estimate the routing policy values $V(\pi_{\text{arm}})$; U_{train} here), and P_{dep} (final selection/routing, reading activations only through the frozen P_{train} operators). **The partition is locked:** every confirmatory estimand is conditional on it (Eq. (12)). Y_{obs} and U_{train} never co-occur in an equation; no P_{dep} outcome enters the P_{dep} decision. A multi-pool re-split is a separate *non-confirmatory* generalization probe with its own Holm accounting.

Baselines and routing arms (locked). Confirmatory selection comparators: `nait_layerwise` over $\mathcal{L}$ (the stronger, decisive NAIT), `endpoint_neuron_selection` (ϕ_{end} , full control), `random_subset`, `full_data_it` (upper bound), `quality_score_selection`, `influence_gradient_selection` / LESS [39], `zero_shot_select`. Routing arms (R1, all over $\mathcal{A}$ with shared r_0 and shared `capacity_match`): `uniform_lora`, `shuffled_psi`, `random_mask`, `global_surgical` [27], AdaLoRA [44], optional NeFT [41], `lora_moe_routing`. The 8-anchor NAIT is a secondary diagnostic, never the headline comparator. “Beats NAIT” requires beating the stronger full- $\mathcal{L}$ variant; “beats SOTA selection” requires beating *every* comparator under the IUT.

Multiplicity (closed-testing graph, FWER $\alpha = 0.05$). The confirmatory family is the six nulls $\{R0, R1, G2t, G2r, G2h, G2c\}$. The graphical scheme (Maurer–Bretz) [3] routes all mass through R0: $R0 \rightarrow R1 \rightarrow G2t \rightarrow \{G2r, G2h, G2c\}$ with locked weights ($w_0=1$, edges 1.0, split $(w_r, w_h, w_c)=(0.34, 0.33, 0.33)$, recycle 1.0). R1 and G2t are *single union-null leaves* decided by the full-union IUT (Eq. (13)); the within-trajectory R0 node uses two-layer Holm with the joint block as gatekeeper. FWER ≤ 0.05 is established in-document (Theorem 3). Ablations (D-only, κ -only, J on/off, $\mu=0$, Δ_ℓ^{LOL} , the multi-pool probe) are *exploratory*, outside the confirmatory α .

Precondition gates. *Factuality (G6):* λ_f may be non-zero only if, on a held-out slice, Spearman $\rho(\hat{f}, \text{drift}) \geq 0.3$ and Pearson $r \geq 0.3$ (one-sided lower 95% CI > 0) and ECE ≤ 0.1 [18]; else $\lambda_f := 0$. *Outcome reliability (G7):* before the R0 regression on the influence proxy Y_{obs} , require ICC(2, 1) ≥ 0.6 across ≥ 3 seeds [36] and Spearman $\rho(Y_{\text{obs}}, \Delta_{\text{retrain}}) \geq 0.3$ (lower 95% CI > 0); else fall back to the direct retrain-delta and do not run the proxy regression. *J-validation:* $J_{c,\ell}$ enters the router prior only if it correlates with Δ_ℓ^{LOL} ; else dropped.

Measured compute ledger (cost_model.py). Per arm: parameter count, optimizer-state slots, *realized* FLOPs (measured, not rank-inferred), wall-clock, batch/accumulation policy, and a skip-flag (whether masked adapters are excluded from the backward pass). R2’s matched budget requires equality within a locked tolerance on params, optimizer slots, and realized FLOPs; the shared r_0 substrate and shared `capacity_match` reduce the match to $\mathcal{A}$. Equal rank is *not* equal compute; an extraction-parity multiplier $> 2.0\times$ declares high-cost-analysis with no deployment claim.

Locked symbols, thresholds, and seeds (pre-registered design choices). All pre-registered *decision* thresholds are **locked now** in `configs/experiments/lattice_v5.yaml`; they are design choices fixed before any run, *not* results, so the analysis cannot be tuned post hoc. Only genuinely empirical quantities (resolved checkpoint SHAs, measured FLOPs, observed gaps) remain `DATA_NEEDED`. Shared seeds 0..19 (`configs/seeds/paper_20.txt`); 10-fold family-stratified cross-fit for R0; $S \geq 20$ seeds for R1/R2; $B_{\text{bt}}=10000$ bootstrap- t resamples. *Capacity (feasibility-coupled):* $R_{\text{tot}}=64$ allocated over $\mathcal{A}$ ($|\mathcal{A}|=8$, so an even split is the standard rank-8 per anchor), $r_{\text{max}}=16$, $r_0=4$ on $\mathcal{L} \setminus \mathcal{A}$. r_{max} is pinned so $|\mathcal{A}| \cdot r_{\text{max}}=128 \geq 64=R_{\text{tot}}$, hence $1_{\mathcal{A}} \in \text{DOM}$, $k_{\text{min}}=\lceil R_{\text{tot}}/r_{\text{max}} \rceil=4 \leq |\mathcal{A}|$, and `capacity_match` is total and score-free (Lemma 2). *Margins (normalized [0, 1] utility):* $\delta_{R1}=\delta_{\text{target}}=0.01$ (one normalized utility point—a *meaningful* routing/method win). The margin is a power-justified *design* floor, not a fitted constant; the back-of-envelope derivation is: with $S=20$ paired seeds and a target per-seed paired-difference SD $\sigma_d \approx 0.02$ on normalized utility, the standard error of the mean gap is $\text{se} = \sigma_d/\sqrt{S} \approx 0.02/\sqrt{20} \approx 0.0045$. At a plausible true gap $g \approx 0.025$ (just over $1\sigma_d$ of per-seed separation), the non-centrality is $(g - \delta_{R1})/\text{se} = (0.025 - 0.01)/0.0045 \approx 3.3$, against a one-sided $z_{0.95} \approx 1.64$ critical value, giving $\Phi(3.3 - 1.64) = \Phi(1.66) \approx 0.95$ asymptotic power, conservatively $\gtrsim 0.8$ once the finite- S bootstrap- t

Table 1: Implemented v5 modules (do-not-run, pure-Python; green harness, no run executed). Status is *implemented & unit-tested*, *not* executed.

Module	Method block	Status
layer_function.py	capacity_match/make_feasible_mask/controls (Sec. 3.4)	implemented
cost_model.py	measured compute ledger, extraction parity (Gate R3)	implemented
baselines/nait_layerwise.py	faithful NAIT over $\mathcal{L}$ (Eq. (2))	implemented
analysis/layer_attribution.py	frozen $\mathbf{B}_\lambda$, out-of-sample $\tilde{\mathbf{T}}$, ψ , coupling	implemented
analysis/routing_intervention.py	policy value, bootstrap- t , IUT (Gate R1)	implemented
analysis/matched_budget.py	R2-target IUT + margin sub-claims (Gate R2)	implemented
analysis/closed_testing.py	graph, IUT leaves, 63-intersection closure, FWER	implemented
analysis/pool_firewall.py	locked three-pool firewall, leakage assertions	implemented
schemas_v2.py (RouterOutputs, provenance)	auditable records (additive)	implemented

quantile inflation is accounted for. A smaller margin would not be practically meaningful; a larger one would demand implausibly large true gaps to clear at $S=20$. $\delta_{\text{rel}}=0$ (non-negativity guard atop δ_{target}), non-inferiority ceilings $\delta_{\text{ret}}=\delta_{\text{hall}}=0.02$ and $\delta_{\text{cost}}=0.05$ (tolerable drift/cost). *Floors/gates*: partial- R^2 floor $\text{floor_partial}=0.01$ (the residual must explain $\geq 1\%$ of residual variance beyond $[\phi_{\text{end}}, \text{NAIT}(\mathcal{L})]$), $\rho_J=0.30$ (moderate monotone agreement), τ_{sel} the P_{train} quantile yielding mean raw anchor cardinality $\bar{k}=4$. *Control specifics*: fixed seeds $\text{seed}_{\text{rand}}$, $\text{seed}_{\text{shuf}}$, seed_{ada} (integers outside 0..19), the surgical block A_{glob} as the top-4 anchors by pooled $|\psi_\ell|$, AdaLoRA warm-up $W_{\text{ada}}=200$ steps. *Pooling*: pool_split_seed fixes the single locked $(P_{\text{train}}, P_{\text{val}}, P_{\text{dep}})$ partition; marginal level $\alpha=0.05$; ledger tolerance 2%; the λ grid is cross-validated per fold. The closed-test α -allocation ($w_0=1$ on R0, unit edges $R0 \rightarrow R1 \rightarrow G2t$, split (0.34, 0.33, 0.33), recycle 1.0) is the pinned graphical scheme (Sec. 4).

Implementation status and the green unit harness. The v5 modules *are implemented* as additive, do-not-run, pure-Python (stdlib only; no model load, no server call, no training); callers supply already-measured tensors/contrasts at run-later. They are exercised by a green harness of **93 passing** formula-evaluation tests across six files: `test_layer_routing.py` (26), `test_trajectory_selection.py` (35), `test_residualize_gates.py` (19), `test_endpoint_baseline.py` (7), `test_selection_schema_metrics.py` (3), and `test_project_contracts.py` (3). These cover the coupling identity in/out of sample, `capacity_match` conservation/score-freeness across all feasible cardinalities, the IUT size at the least-favorable configuration with the margin entering the statistic, deterministic control maps (including π_{ada} frozen outside V), the faithful-NAIT-over- $\mathcal{L}$ reproduction, the firewall leakage assertions, the closed-testing shortcut=brute-force equivalence with FWER control, the R0 permutation placebo level, and the Brier-propriety/G6 checks. `server.authorized` stays **false**; *no run is executed*; these are code-correctness checks, *not* evidence. Table 1 maps modules to method blocks.

6 Results

This is a Stage-1 Registered Report: no experimental numbers are reported. The tables and figures below are pre-registered placeholders describing exactly what will be reported after the authorized run; **no value is fabricated**. The named primary cells are: R0 = Qwen2.5-7B (math) partial- R_T^2 decision; R1 = the IUT verdict over the five controls at δ_{R1} with $L_{R1} = \min_c L_c(\alpha)$; R2 = the retention-adjusted-gain win vs. `nait_layerwise`($\mathcal{L}$) under the matched ledger.

Table 2: **[DATA_NEEDED]** *Gate R0 — mechanism certificate*. Per backbone/family: median-of-folds partial- R_T^2 and $\Delta R_{\text{overall}}^2$ under the full $[\phi_{\text{end}}, \text{NAIT}(\mathcal{L})]$ control, conditional-null block-permutation p ($P=5000$), cluster BCa 95% CI ($B=2000$), decision vs. the locked floor.

Backbone / family	partial- R_T^2	$\Delta R_{\text{overall}}^2$	perm. p	BCa 95%
Qwen2.5-7B (math)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
Qwen2.5-7B (code)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
Qwen2.5-7B (QA)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
Qwen2.5-1.5B (all)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]

Table 3: **[DATA_NEEDED]** *Gate R1 — interventional routing certificate (the headline mechanism)*. Per-control pool-conditional gap $\hat{g}_c$, paired bootstrap- t lower bound $L_c(\alpha)$ vs. δ_{R1} , and per-control reject; the IUT rejects iff every $L_c(\alpha) > \delta_{R1}$, with the simultaneous bound $L_{R1} = \min_c L_c(\alpha)$ ($S \geq 20$ seeds, compute-matched).

Control arm	$\hat{g}_c$	$L_c(\alpha)$	vs. δ_{R1}	
π_{unif} (uniform LoRA)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
π_{shuf} (shuffled ψ)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
π_{rand} (random mask)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
π_{global} (surgical)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
π_{ada} (AdaLoRA)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
IUT verdict ($L_{R1} > \delta_{R1}$)		[DATA_NEEDED]		

7 Limitations

1. **The residual attribution may be null (R0).** If $[\phi_{\text{end}}, \text{NAIT}(\mathcal{L})]$ already explains the variance, R0 does not reject and the novelty claim stops. This is the intended falsifier; the honest prior is that the residual may be small.
2. **Routing may not beat uniform/AdaLoRA (R1 null).** R1 is a pre-registered policy-value kill-gate with a clean fallback (drop routing, keep selection-only). Publishable either way.
3. **ψ is predictive, not causal.** ψ only *proposes* the policy; the policy value $V(\pi_\psi)$ (real runs, once executed) would certify it. Ridge/fold instability that moves ψ mass without changing prediction fails the π_{shuf} arm.
4. **No per-example routing effect is claimed.** The estimand is a run-level policy value; the only per-layer notion is the descriptive Δ_ℓ^{LOL} , labeled non-causal and outside the confirmatory α .
5. **Scope is the locked pools.** $V(\pi)$ is the pool-conditional seed-expectation; the pool-marginal statement requires the separate, non-confirmatory multi-pool probe with its own multiplicity.
6. **Per-contrast inference is asymptotic.** We claim only the asymptotic validity of the paired studentized bootstrap- t ; the exact-finite- S sign-flip claim is withdrawn, and the one route to exactness (an actual seed-to-arm randomization) is not adopted.
7. **Compute matching is fragile.** The ledger requires *measured* FLOP/wall-clock equality and a skip-flag; unmatched arms are diagnostic-labeled and “savings” is forbidden without a measured reduction. Equal rank is not equal compute.

Table 4: **[DATA_NEEDED]** *Gate R2 — compute-matched method win vs. NAIT and the baseline set.* Relative target gain, retention-adjusted gain, retention drift, hallucination drift, and the ledger-match decision over ≥ 20 seeds; R2-target is the IUT over the baseline set (δ_{target}); each drift/cost sub-claim is a one-sided margin test.

Selector	rel. gain	ret.-adj. gain	ret. drift	hall. drift
random_subset	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
influence_gradient / LESS	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
NAIT (nait_layerwise, $\mathcal{L}$)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
LATTICE-R (ours)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]
full_data_it (upper bound)	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]	[DATA_NEEDED]

Table 5: **[DATA_NEEDED]** *Honest cost ledger (Gate R3) and precondition gates G6/G7.* Per-pool selection multiplier; differential trajectory extraction/storage I/O; amortized validation ($R=1$, break-even R^*); the extraction-parity decision vs. $2.0\times$; and the realized G6 (Spearman/Pearson/ECE) and G7 (ICC, proxy $\leftrightarrow$ retrain ρ) values.

Line	value	gate decision
per-pool selection multiplier	[DATA_NEEDED]	[DATA_NEEDED]
differential trajectory extraction/storage	[DATA_NEEDED]	[DATA_NEEDED]
amortized validation ($R=1$, R^*)	[DATA_NEEDED]	[DATA_NEEDED]
G6 (Spearman/Pearson/ECE)	[DATA_NEEDED]	[DATA_NEEDED]
G7 (ICC, proxy $\leftrightarrow$ retrain ρ)	[DATA_NEEDED]	[DATA_NEEDED]

8. **Off-manifold J .** The layer-ablation profile is validated-or-dropped against a real anchor layer-freeze policy-value difference.
9. **Factuality proxy validity is a precondition, not a result.** Brier proves calibration, not eval-drift prediction; fail G6 $\Rightarrow \lambda_f := 0$.
10. **Influence approximation.** First-order LOCI influence can be inaccurate for large moves; G7 is the explicit precondition—fail $\Rightarrow$ no primary claim on the proxy.
11. **No empirical claim yet.** Every result is design; the only numeric checks are the green-harness formula evaluations, which are code-correctness, not evidence.

8 Conclusion

We presented LATTICE-R, a Stage-1 Registered Report that turns the per-layer axis NAIT computes-and-discards into a decision variable: one endpoint+NAIT-residualized trajectory attribution drives both data selection (its sum) and LoRA layer routing (its support), tied by an exact coupling identity. We make the routing object a well-posed, fully-specified *pool-conditional policy value* over whole training runs—no per-example potential outcome, no SUTVA, a score-free closed-form capacity map, and deterministic controls—and the design *would certify* the coupling, once the authorized run executes, with an interventional routing test (six real masked-LoRA arms, an intersection-union test over a paired bootstrap- t). A ridge-FWL mechanism certificate residualized against NAIT over the full $\mathcal{L}$, a measured compute ledger, a locked three-pool firewall, and a closed-testing

[DATA_NEEDED] *Figure placeholder.* (i) The R0 permutation null of the cross-fit partial- R_T^2 with the observed statistic and BCa CI marked; (ii) the R1 forest plot of per-control bootstrap- t lower bounds $L_c(\alpha)$ against δ_{R1} with the binding control highlighted. Rendered after the authorized run.

Figure 1: **[DATA_NEEDED]** *R0/R1 inference visualizations* (no data at Stage-1).

graph with an in-document FWER bound complete the plan. The theory is discharged where provable and the modules are implemented as do-not-run pure-Python with a green 93-test harness (`server.authorized=false`; no run executed). **Empirical validation follows the authorized run**; no experimental numbers are claimed here, and a null at any gate is a clean, publishable falsification.

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